ActInf Textbook Group ~ Cohort 9 ~ Session 5 (Chapter 4) ~ 9/18/2025

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Cool. Okay. So, we've started recording. Uh I wanted to get started now. Uh thanks everyone for for joining. Uh September 18th. Uh today we're we're still looking at the first half of the active inference textbook from 2022. Uh today is for chapter 4, but we can take the conversation wherever we like or if anyone has anything they're interested in in particular. Um, but I did want to touch a little bit on chapter 4 just for the sake of the the scheduling and kind of connecting it with with previous chapters which is that um chapter one we had like a very loose general introduction act of inference as a uh first and foremost is it is a framework. It's highly um related to and entails the uh free energy principle, the idea that uh everything the brain does must minimize free energy. Um we're we're introduced in chapter two to the low road to active inference. So we're talking about kind of basian frameworks using kind of basian statistics or mechanics. Um it is still you know a topic for debate. um for those who adhere to something like a basian brain hypothesis whether they think that it is expressly a basian sort of framework that the brain itself follows or if it does something like that. Um there are kind of representationalist versus non-representationalist accounts of what the brain does. Um the the the core aspect here is that something like basian statistics and mechanics are involved. Uh most put most generally is the idea that whenever we come to a situation, we have priors prior sort of evidence um that we bring to a situation when confronting new observations when eliciting new acts um actions when updating our beliefs along the way. And uh so then chapter 3 is the from sort of the the mountain top of the high road looking down which is the claim that um it essentially begins with the the free energy principle and talking about why is it that organisms that that entities things people um brains or otherwise how do they continue to exist uh and persist um how do they stay alive when they are living? Um and and it's it's there that the free energy principle comes in as that is how they do that and why they do that. Why they continue to persist is through this this kind of repeated continuous engagement with free energy minimization um in order to sustain oneself in order for all necessary parts to sort of self-organize the the sort of guiding principle is is free energy minimization based upon you know several decades of work that that Carl Fristen and others to Par and Zulo and and um Isaura and and others have have have kind of empirically established in various ways uh through neuroiming neuroscience kind of related studies um working with cells and cell cultures up to neuroiming data in the human brain or the pavalian brain and um and then it's here in chapter 4 that things kind of come back around and we start getting a bit more into the the the sort of like here's how we put things together these basian frameworks Free energy. What is free energy? How does that how does that work and how does it function? Active inference for better or worse is slightly notorious for having relatively complex um kind of equations and formalisms involved. So it's it's really in chapter 4 that that we were exposed more deeply to to a lot of the aspects of that. Um already uh in the in the first section we return to Baz's rule where we know that we have a prior or a prior belief about hidden states. We have our our marginal likelihood or our evidence which is that probability of y which is kind of like the probability of observations. Um and then uh the conditional probabilities of each on each which would be the posterior which is the probability of X conditioned on Y. It's kind of what what do we believe given the observations we we have received right conditioned on these observations this evidence this this this model evidence this data um and then the the the flip of that which is the likelihood which is what is the probability of this observation conditioned on what I believe to be the case what I believe the hidden states of the world currently are right and it's with those four components and using bases role to kind of move and compute between them. Um that that we kind of proceed and then from there the issue has to do with computational tractability. Um, you can imagine for someone I I I attempt to use kind of everyday examples of these things, but for for someone who decides that they want to go to a particular location, say they want to go to a store, um, they want to buy groceries, they'll they'll get they'll they'll purchase food, they'll gain their sustenance, you know, for the day or for the week or the next few weeks or otherwise, um, you know, they'll come home and they'll maintain their homeostasis. Um if if this person is say I don't know in Par lives in Paris and they're the store is a few blocks away in Paris then the idea that they need to run some kind of broadscale inference over all known locations that they're aware of on the planet whether that be in the United States or in you know um China Russia or otherwise um that that it's we intuitively we would think why would you need to think of all these other places. Similarly here in a certain sense it's like well if you had to kind of integrate or sum over that level of vastness in a in a computationally speaking in that vast of a state space um things would become quickly intractable uh inefficient very very energy and timeconsuming um we we already know that there'd be a lot of unnecessary computation involved but that that's not the point right it's it's like this is based upon how the the brain works, whether it has that additional perspective on is it efficient for me to think about what's going on in um in the United States in this case, right? Um so, so to speak a little bit more technically, the idea is that we use something called variational basian inference here. Um where the the authors claim that they they kind of exploit what's known as Jensen's inequality. It's this idea that uh and it's not just an idea and there's kind of an illustrated proof of this but that the idea that a log of an expectation is always greater than or equal to the expectation of a log. It's there that they kind of do this um this trick without meaning to to to um make it sound like an informal thing. It's very reasonable thing to do in variational methods. um but to actually like have this kind of variational posterior that allows for variety of other computations. So they talk about like mean field approximation and making different kinds of assumptions of how things work as sort of these shortcuts um that allow one to or allow a brain or allow some kind of um process to to navigate this otherwise intractable state space that it's confronted with. perhaps composed of. Um, all this is to say that this one is much more biologically plausible. It allows for focusing more on what is more significant in the moment. Allows for marginalizing out from massive state spaces like we only need to focus on Paris because that is where the store is. That's sort of that sort of logic. Another term that they they use here is importance sampling. Right? So being able to to kind of hone in on the the what what's more salient um in the moment actually having the kind of mathematical architecture that allows for you know someone who wants to do active inference or other forms of uh research and computational modeling like that gives us a means to ourselves like write this down uh write down the formalisms or the equations right um and being able to to make things more ractive. So um so on page 66 we have an equation a more express equation for v for variational free energy which is uh divergence minus log model evidence we know from the >> Andrew do you want to share your screen? >> Uh oh yeah am I am I not sharing my screen? >> Uh the symposium page >> it's still the symposium. Yeah. Flip that over here. X64.4. Here we are. Yeah. So this is this is uh one way of formalizing variational free energy. Um it depends on what everyone wants to discuss today. Sometimes I go a little too far into these these like uh chapter summaries. Um so I'll just more quickly say like we can we can we can remember many of these pieces from previous chapters, right? We know that X is uh representative of hidden states, hidden states of the world. Y are our sensory evidence or observations that we receive. So whenever we see this probability of X conditioned on Y, we already remember like oh these are posteriors. We see probability of Y that is the evidence. This Q of X is the interesting piece that comes in through Jensen's inequality and getting to the variational methods. rather than a P, you have a Q. Recall Daniel saying, "Mind your P's and Q's in the past to to kind of remember that." Um, but Q is that variational um Q is that variational posterior, right? It's just to represent like now now we're in the realm of using variational methods or that it's it's this Q that we use to denote that um out of sort of remembrance and um and then from there we get into like here are a variety of different sorts of models that where we could where we could apply um these kinds of methods. Some of them may not actually call for variational methods. Um, it depends on the complexity of the problem at hand. But as far as free energy minimization goes, you would be using variational methods regardless of the circumstance. Um, so these models for example, um, for anyone unfamiliar with doing any kind of modeling in general, a really good example is just this one in the top left where we do have a prior, we do have a likelihood which is the probability of the observation conditioned on uh, in the states of the world. And um, a really good example would be uh, a COVID test, right? Um we like we some would like to believe that a COVID test could one to one tell you if if you have COVID or not, right? But at the end of the day, ultimately there is an accuracy rate because what's really going on with a co COVID test is a kind of modeling paradigm that uh can go by the name of binary classification trying to decide do you have COVID, do you not have COVID, right? And so what's going on here to use this as modeling a this is could be a model of a COVID test is the probability of X is kind of like saying what is the probability of having COVID? We don't really know. The test is sort of like your your sensory receptor here, right? it's going to take in some kind of bodily, you know, fluid uh in in some shape or form and then it's going to um run some kind of internal complication as it were uh based on how the the test is designed and give you final reading at the end. So it's it's there that you know based upon whatever is going on in the the the the material that's given to that test there's an inference being made about whether you have COVID or not and that's being returned to you sort of a separate agent to receive that as your own observation. Um, so that's it. It it leads into this division where you have things like true positives and and true negatives and false positives and false negatives that the test told you that you have COVID when in fact you don't and so now you have a false positive. Um, all of those kinds of dynamics are readily captured within this like given given what the test finds within the the bodily fluid given to it. But is uh what is the probability of seeing the that composition of that bodily fluid given you have COVID versus given you don't have COVID, right? Like there it's as if there's a kind of inference being made. We could do the same thing. We could say, "Oh, uh what is the probability of me seeing that this test tells me I have COVID conditioned on me having COVID? If I trust this test, it's probably very high. If I don't trust this test very well, then well, I don't know why I'm using it in the first place, but more important like to the the point here, it'd probably be a much lower probability, right? Um, so that that that's kind of the that's kind of a shorthand of uh, you know, how to how to examine these models and from there it gets simpler as long as you kind of catch on to the first one. This is called factor graph fory factor graph notation. So, um, like this upper right one, it's like, well, now we have two different variables that are going into Y. You know, what what is what is my weight? It could be a function of my my height and my diet, right? And those are the two thing those are the X and the Z that that end up leading to a Y, right? Um or you could have a single variable that's impacting two other variables, right? So how well do I sleep at night and how well do I function during the day are both functions of you know how stressed out I am. So and then you can have these longer kind of running chains right of one thing that impacts another thing that in turn impacts a third and that that can start to expand outward further. So and then these models here are in many ways similar to the previous ones although granted they're more complex. These are only two models and there many more circles and squares to look at. Um but the main idea is that we're moving in the direction of what we'll see in chapter seven and 8 which is kind of these more fullyfledged predictive coding and active inference models. Um both of these are formally equivalent. They just use different symbols. Uh so in the top one you have like pi, you have o, you have s. the bottom you have V, you have Y, and you have X. Um, the only reason why they do that is that this top one is for what's called discrete time models. You can think of discrete in the sense of colors, not as like a RGB like color scale, but just red versus blue versus green, like these discreet specific things or am I hungry? Yes or no. Right? as opposed to the bottom model which would be used for continuous time which is where you start to look more at scales. You look more at things like continuous continu continuously valued metrics. So heart rate um um signals from an EEG electrode, you know, sensor in neuroiming studies. um the amount of money in your wallet potentially if you bring it down to the cents. Um which could also be due to discrete sense of how many pennies do you have but uh so the main thing is that we have discrete time and continuous time. There's some nice literature elsewhere that shows that they kind of they both at at their root exhibit the same kind of behavior virtually. So they both reduce back down to the free energy principle in a sense. Um which is really nice. It keeps the framework very consistent. Um and yet you can use either one depending on the situation again if you're trying to model something. We only focus on the top one. And I'm already going into this more than I need to because it'll all come back in chapters seven and eight. Um but this is kind of where we start getting to the idea of oh from left to right there's a temporal horizon that is these are the these are about beliefs about h how hidden states change over time um and then reading it top to down we can see how an o observation is taken in s a hidden state is inferred and then inferring a hidden state of the world leads to pi and action or policy to be taken and that all of these things are more or less kind of kind of synchronized with one one another and they proceed processually. So, so you know you take in an observation about the world, you update your beliefs about what is going on in the world given that observation and finally you move on to action or planning or or carrying out some kind of action in response to not directly to the observation but to your intermediary belief about the world that was updated. So my stomach growls. I infer that I'm hungry. I decide or act to go get food to save my hunger or information gain. I I start looking at restaurants where to get some food or something. Check the fridge. Do I have any food? Then so there there are plenty of other topics going on in here, but I already know that that someone wanted to to to discuss um potentially other parts of the book or other parts of this chapter. Um so I'll just wrap up that there will be more of these kinds of things in chapter five that's much more focused on what's called message passing um and kind of the neurobiological aspects. We get discussion of sort of um deep versus superficial pyramidal cells um in in in cortical microircuitry in the brain. Um and and so a lot of these things just have to do with passing prediction error uh and and updating beliefs accordingly kind of at this this sort of more micro level as it were. But that's it's there that we can start to look at how these things actually map to biological coralates and are not just sort of free floating mathematical ideas without any empirical grounding. Generalized coordinates of motion. these play out in the continuous time models that I mentioned earlier. Um that these are especially they're they're very important for understanding continuous time models. Um but also they're very illustrative whenever it comes to the idea of you know reflex arcs and and motor movements. So, you know, how is it that that through a discrete decision in my brain to, you know, grab uh my cup of coffee near me that my arm actually like is able to navigate itself to my cup of coffee, right? Um the trajectory, the kind of physics involved in that so on. Of course, that that kind of computation can play out through variety of other um examples, not just movement, but I think movement is a quick one of kind of getting at it. Here again, we have more of these. It's really just showing conceiving of this as a kind of message passing, right? a message passing between different neuronal populations in the brain or collections of neurons or potentially different brain regions depending on what level of ex abstraction you want to move to uh whenever you do you're doing this kind of modeling or trying to understand um a process. So I'll wrap up there. Um yeah. Um I think a team were you interested in um you mentioned something about another chapter or if there was anything you're considering. >> Yeah, I had a a question about uh about I think this this comes back in chapter 4 about the pragmatic and epistemic value or I don't know about the exploitation exploration trade-off. I was kind of curious as to how this is uh how the how an organism or an agent uh trades off these two these two modes of uh of of reducing uh of reducing uncertainty of surprisal because it's I mean as I understand it in the short term you want to reduce the the pragmatic value the pragmatic uh prag I forget how it's called in the in the book yeah the pragmatic value. But in the long term, you want to reduce the epistemic value because uh in if you if you make if you if you're in an environment that uh that is unknown, you have to explore and then this exploration will will give you the the ability to to be less surprised in the future. So you you kind of trade off the the the short-term surprise for the long-term one. So I'm kind of curious as to how practically this is done like uh uh what kind of uh what kind of uh of uh situations like uh the what kind of scenarios is an organism placed in like the ones that I sketched out is are there other ones that that you can think of like uh where the where the pragmatic value is is better than the epistemic value or vice versa or is it I don't know is or did I cover everything maybe in my in my kind of explaining explain the question but I don't know if it's clear >> everything that you said makes a lot of sense and it's a it's a very significant question here um it is for active inference and and they do kind of make a point about this whenever I've know given um done presentations where I'm describing active inference to others this is actually a point that I I particularly am fond of and it's part of why I uh am interested in active inference as opposed to other frameworks. Um there there are many reasons why I personally like it more than other frameworks. Um but but this is one of those reasons. Uh so here on the the screen I've come to this is equation 2.6 which I would infer is probably in chapter 2. I can't remember off the top of my head, but they usually follow that numbering. Um, and we'll just I this is quite a monster of an equation because it is technically a singular full equation. Um, but you can just read this top line with information gain and pragmatic value. Um, which is those those terms that you were referring to. Um and they do for better or worse there are a lot of different words that are used. So sometimes information gain is called epistemic value just to have another value that can be kind of more readily like coupled to the idea of pragmatic value. Um information gain works as well. Um and and then at the second line you see this is just an equals sign. So these are actually one way of doing expected free energy which is G. You'll notice that that's different from F. Um but and that that's an important distinction. But fortunately that's the only distinction we need to make. The only two forms of var of of free energy that we're looking at in any of these models are variational free energy and expected free energy. What is expected free energy? It's incredibly similar to variational free energy. It's still a variational basian uh inference method. Um many of the formalisms are the same. You're still working with a prior and a likelihood and all the rest. The only difference is that variational free energy is sort of like a reactivity to the moment. An expected free energy is what's related in what we would call planning. uh you know having a kind of policy horizon where you you know if if I infer right now that I'm you know I receive an observation my stomach is growling I'm surprised by that I didn't think I was hungry but now I'm starting to think I must be there's some kind of surprisal that happens um you know my my variational free energy of that moment so to speak would probably uh go up so remember the idea is we want to minimize that as uh and that's what chapter 4 in part is about is that variational free energy is a an upper bound on surprisal. So by minimizing variational free energy, you're necessarily going to be um approximately minimizing surprisal. So so back to that kind of funny example I'm giving. Um you know there's surprisal in the moment. Oh no, I'm hungry. Now I start planning. What do I do? Right? How do I we we would call this in technical terms like policy inference. I need I I I infer a state of the world is that I'm hungry. Now I need to infer what policy to take. That's why we have these two different equations, right? But they're they're they're essentially both variational methods. It's just expected free energy is involved in planning. And it's kind of it I I I can't remember if it's Fristen or or someone else, but someone has phrased this as sort of the free energy of the future. um like what what free energy do you expect uh given you do this action which could lead to this change in the world what kind of v what level of free energy do you expect from that right and so these kinds of computations could could basically be carried out into the future up to some horizon um you know if are you thinking about your life in the long term and where you'll be in 10 years from now are you thinking more about going to get food uh or or are you thinking about making plans with a friend next week where there's a lot of things that could happen between now and next week. Um in any case, you're you're you're thinking into the future and it's there that we sort of depart from simply complexity minus accuracy, a variational free energy of the moment and instead we start thinking going forward. And so now to get a little bit more deeply into kind of the way you you phrase the question of looking at information gain again otherwise known as empymic value and pragmatic value. One way of viewing these in kind of simple terms is information gain means actions that will help you learn more uh about your environment. Uh really more specifically, it's saying you want to maximize the difference between your beliefs about the world given observations and actions versus your beliefs about the world given actions. Like regardless of whatever it is you're saying. So, and to this day, it's still been hard for me to come up with a fully intuitive example of what that means. But, but the main thing you'll notice between these two cues, uh, is the why. The why is present here. It's kind of like saying, well, well, I think that if I do this, then this will be what comes out of that. The the hidden states of the world will change in this way. Oh, if I if I eat this food, I will no longer be hungry. Um, that would be the right term, right? I I I I did something and I'm trying to change a hidden state about my hunger, which I can only infer. Um, the the earlier term is kind of like saying, "Oh, well, I I ate this thing. It turns out it had no caloric value uh and was very small and hardly fills my stomach, so it turns out I'm still hungry." like I don't know if eating this particular food will actually sate me or not, right? Um probably a better example would be the idea of reading a book to find more information, right? Like well, you've never observed that book before, so there's no why in it. um you don't really know what you'll have gained from that but maybe by reading it you'll have more information about the world around you thanks to actually observing the outcome of doing this particular action right another way of viewing it is just if I do something if I don't know what will happen then I don't know if this action is ever going to be useful for for anything right um or maybe the action just may or may not be useful for a particular context um whatever it is, we're going to do it just for the sake of seeing what comes out of it. Right? You think of a child, children do this all the time. They're incredibly uh curious and sometimes they do things whether they should do them or not do them or otherwise um you know just just to see what occurs. And then pragmatic value is quite different. Here we see a a term that we haven't seen yet um but it comes up all over the place in chapter 7 whenever it gets back into the modeling C. And this is this C is is is very important. This is used to denote preferences. Um, there are other words that this goes by, but and and preferences is kind of a loose term that that I think it's kind of a carryover from some economics literature or thinking about rational actors or or or probability theory, what it means to have like um a subjective sense of probability or beliefs. Um, preferences here could mean one's preference for a certain, you know, chocolate or vanilla ice cream or something, but it could also relate to homeostasis, right? So I I prefer to have my and none of this none of the computations I'm talking about here none of this is assumed to necessarily be a conscious process but like in a sense my my brain prefers that that my my blood oxygenation in my in my body be at a certain level or my heart rate be around a certain level right that is what keeps me living or surviving. So preferences in that sense can also be you know depending on how you view that like to some degree could be genetically derived just as much as you know they be learned through experience over time of like oh I actually I tried that chocolate ice cream and I did decide I like it more and I prefer that as a more conscious thing right so it's it's kind of like at various scales but the the main point is that you see that like pragmatic atic value for anyone who's interested or or engages in things like reinforcement learning um and and kind of machine learning models. Um th this would be kind of like a reward function, right? Like I there are certain things that I need to do in order to stay alive or oh there are certain things that I prefer. Um I should go after those. um if I do this particular action is that will I it will I get what I want out of it or what I prefer out of it. Right? So that I mean the the autonomic nervous system does this. It repeatedly tries to regulate the body in different ways depending upon the circumstance at hand and it tends to be attracted towards bringing you back to a nice preferred so to speak regulated state. Um so pragmatic I I I I give all this detail because pragmatic value is is it is like a reward function. It's kind of like an attractor an attracting system that the book gets into that kind of idea more in chapter 8. Um so then finally to bring it together. So, okay, well, information gain and pragmatic value. How do we pick? How do we pick if we should, you know, should should I go straight for some food or should I like look at other kinds of food I've never had before? Or in chapter 7, we have a T-aze example where there's a there's a mouse who can get a reward by going either to one side of the maze or the other, to the left arm or to the right. and it doesn't know which side the cheese is on. Um, it has an option to get an informative cue from its environment. Uh, that will tell it where the cheese is, if it's on the left or right side. Um, the the mouse very well could just guess, go for the left arm or the right arm, or if it gets that cue, which isn't immediately giving it any reward, so to speak. it's not getting the cheese, but it is giving it more information about the environment around it. Maybe it should do that. So, a lot of people use that this T-mase example uh as a kind of kind of like how how do people you know how do how does any agent or any entity or organism make these kinds of decisions or or or balance these kinds of trade-offs. Um with active inference there all potential actions are already scored against each other based on how much information each one of them will give you and the pragmatic value that it will give you. So at the end of the day here we don't have to choose this whole equation tells you everything in one go. You compute information gain for this action. You compute pragmatic value in this action. at the end you get your amount of expected free energy that gets compared to all other actions. So there's a phrase in this textbook they say that that sort of like exploration or information gain and exploitation or pragmatic value or reward seeking uh naturally balance out. They don't balance out with sense of like okay now they're half and half like that. That's not what that means. it just means that they're both included in the equation as it were. They both function there. Um uh an alternative way of saying this is um and this is why like this is part of what brought me to active inference on this point. Um, in in reinforcement learning, uh, people will try and decide on the exploration exploitation dilemma by saying, "Oh, well, we'll just have our agent always go for the reward, but we'll include uh an extra rule. We're going to hardcode some rule in there that tells the agent to instead explore, which means do something random 10% of the time." And that mean that that that'll make sure that the agent learns a little bit more and doesn't just repetitively do the same thing over and over and doesn't get stuck in some kind of suboptimal, you know, solution. Um, that is that is a very kind of like that is the computer programmer's kind of decision to to do that, right? to hardcode this rule that tells you to do something random 10% of the time. Or they can make it as complex as they like, but at the end of the day, they're kind of they're kind of hard coding something in there that just is forced randomness to make sure that you don't always do the thing that you're otherwise hooked up to do. Um, this is kind of like saying, oh no, you you are also hooked up to do things to get info. you are kind of hooked up to explore it and it's not purely random like these are all components of the model that already exist because this is a basian framework which means to say what is going to give you the most information is going to be based upon your prior beliefs right so if you want to read a book to gain more information um say you're a researcher in a particular you know established field of science science or something. It's very unlikely that you're going to like read some random fully random book from the library. It's more likely that you'll end up looking for books that relate to your field of interest or maybe you're working on some kind of, you know, project or paper or dissertation or otherwise. And so you find a a book that seems to most match, you know, kind of the direction that you're going for so that you can gain information. So, so it's more like those kinds of dynamics are what are involved uh in exploration under the active inference framework. Um, intuitively to me they make a lot more sense. Not saying that we should always rely upon intuition and understanding these things, but but it was here that I was like, "Oh, like that that actually makes a lot more sense if you were to formalize the idea of exploration, not as pure randomness, but rather you're coming to the situation with what you already know uh so far or believe so far about the hidden states of the world and actions and what you've seen in the past, what might happen in the future. Um, so that's kind of what is meant through a natural balancing of of exploration, exploitation. I kind of turned that into a lecture though, so I want to apologize for that. Did that kind of get at kind of what you were asking team or do you have any follow-ups on that you're curious about? >> Was it was pretty clear and exhaustive, but yeah, if I think of something, I I'll just I'll ask you later, but for now, it's quite good. I mean I mean I it's kind of I mean I'm just processing it right now that you said that it's kind of baked into the model and I'm trying to understand how how this is uh like because it's such a big thing where in reinforcement learning they talk about the exploitation exploration trade-off and I'm trying to see how how what you said right now is that it's basically it basically solves the trade-off is what I understand. >> Yeah. Yeah. It's like what the way to like whatever exploration an agent gets into um it'll be based on its preceding beliefs, right? It'll it'll it'll and at the end of the day it'll be whatever it is whatever it does to get more information will be all tied back to minimizing free energy. Right? That's sort of the the broad simp like despite all the very complex looking equations which the rest of these are just different ways of breaking down expected free energy. Right? So these are just other ways of doing that that all equate to one another or kind of like upper bounds of one another. Um, this is the one I usually come back to and most people come back to is information gained, pragmatic value or it's exploration, exploitation. But yeah, so there's no there's no like do something random here. Um, unless unless the agent had a belief about what it means to do something random, I guess that would technically be a component in the model. Um, but it it would depend on the the agents beliefs at the time. If it thinks it should genuinely do something random, my bet is it probably won't. Um, most of the time it'll do something that gives more context to other things that the agent does care about, right? Um, and both of these are involved. So again, it's like I'm doing a project on psychiatry. I will probably go go to the psychiatry section of the library to read books that directly relate to the pragmatic value to be attained from like continuing my research. Right? So that their info gain and reward or pragmatic value are both involved, right? So that's you know it's not not so much a tradeoff as it is a kind of um summation of these values. Um yeah. Uh, Tara, you have your hand up. >> Yeah. No, I um uh thanks for all of this background and and I feel like you're getting towards some of the questions that I've been thinking about. Um, one of the things is just um, uh, I was listening to one of the lectures and someone made a statement about um, how deep is your generative model and um, I loved that that quote and so I've been would love to to hear more about how how our beliefs are encoded in some of these equations. I I think I just heard you say that um you know and that that the preferences um are sort of in that C for the pragmatic value and then um and then you gestured towards the the why as as the place where we might encode some of these um uh beliefs. But um you know trying to and and also thinking about how beliefs get encoded um in terms of time scale you know like are we are we encoding the beliefs of the experiences I've had over the past week or am I encoding the experiences that I have over the course of generations you know so like this um and and and and then the question of to what extent they're in hidden states um you know, I'm I'm sort of aware of my thinking, or at least I'd like to believe my thinking over the past week, but I may not be as aware of some of the things I've inherited over the course of, you know, generations. Um, and that that I'm I'm interacting with both of those sets of beliefs. Anyway, >> sure. >> Yeah. No, good good uh thanks for that and good question. It's um there there are a few different things I could say. uh with so with the idea of encoding beliefs or at least how we will typically represent them whenever we you know try our hand at act like you know making a model ourself and putting it into sort of a computational program to to to sort of model you know an organism a mouse in the teams or whatever. Um most frequently we we're working with um at least it it kind of the case everywhere but we we'll go with the discrete time models like as I mentioned earlier um those are usually done in I'm trying to see if we um I'm trying to remember if this is a figure and equation we're usually working with matrices um might be this is chapter Okay. So, they'll look something like this. And this is back to the the teammates's example. so this this is a little out out of context, but this this these matrices are all part of um what an overall established B matrix, which is kind of the common nomenclature for better or worse. Active inference kind of does this thing where we start from A and then we go to B and then we go to C and there's no inherent meaning for why it's a B. Um it's used to denote what's called a state transitions matrix. And so this would effectively be an agent's beliefs about how hidden states change over time conditioned on what they do like conditioned on policies. Honestly, out of all of the different components, it's it's the most it's the most complex one. Uh because it involves both time and it also involves action. Um so, uh I'm currently hungry. I believe with a probability of one that if I eat something, I will no longer be hungry. Right? So, that's it's like a a hidden state of the world right now, an action I can do right now, which is to eat. in the next proverbial time step uh I will no longer be hungry like that by by me doing that I have changed the state of the world and that so th this this is directly you know a part of action whenever we're talking about action and perception in active inferex and elsewhere um so this is kind of how they would they would look right um is that the these rows and columns are like saying like oh if the agent well I'd have to break this down further but But each one of these has to do with an action and has to do with the agent moving uh through through the T-maze. So it believes with the probability of one that if it starts at the the center and it chooses to not move then it will remain at the center or it uh alternatively if it starts at the center it could believe with a probability of one that if it goes left then it will end up being uh at at the left arm of the the maze. Right? So the the um these are the ways of kind of attempting to simpl simplify that kind of logic and in like verbally. Um but all of this is to say that these are just effectively a lot of matrices with their own respective kind of probability values. Um and then whenever it comes to real time inference all of these are playing into decision making. uh free energy is kind of just um in one sense a kind of computation over a bunch of matrices uh trying to find what combination is needed to achieve the lowest value of of of free energy. Um that's that's what it all comes back down to. Um, and you can imagine if we used all of these matrices, every single bit of them fully, and tried to fully compute surprisal, um, that it would be a lot. And so it would be too much and it would be very far away from something like like biological plausibility. Like the human brain runs effectively on the calories equivalent to I think two bananas um, every day. Um it it would be incredible if if we did a massive full computation over everything with with billions of neurons at any given given moment and only need to eat so much food to survive. So um I hope that that was kind of kind of gave some perspective on those things about this encoding and then with actually yeah with um with with temporal horizon which is another good question. Um, so not only can you have kind of these state transitions like I was referring to, um, as in like, oh, if I go left, I will then be in the left square. If I go right, I will, uh, I'll be in the right part of the maze, excuse me. Um but you could also have these kinds of active infer not just active inference but the brain generally this is this is very broad throughout the literature whether you're kind of doing a more fristenoriented um um you know perspective on these things or not is that the brain tends to work kind of hierarchically there are interacting systems between one another right and they can all kind of operate at their own respective time scales So, I I'm someone who who's done some work and and we'll be presenting of it some of it next month uh on um u post-traumatic stress disorder. And we're trying to computationally model some dynamics that are involved in that. Not not the experience of a traumatic event, but rather what comes after that, someone's capacity for kind of like navigating the world around them. So, a lot of that has had to do with looking at the differences between whenever someone with PTSD or anyone regardless of whether you've been diagnosed with that or not. Um, whenever you come into contact with with a phenomena, you have a certain part of your brain, a region of your brain, including what's called the amygdala, which is very responsible for helping with threat detection. Um, other things too, reward detection. In fact, it it works at a very fast pace. Uh whenever whenever you're you're engaging with the environment around you, it actually it it tends to activate and then send information up, you could say perhaps hierarchically to the prefrontal cortex, which is the area that we most commonly associate with, you know, consciousness, rationality, deliberate decision making, not impulsivity. Um, so it works at a slower pace than the amygdala does, right? Um, an even slower functioning part of the brain, the hippocampus, is where so much of your long-term memory is stored, including things from that happened whenever you were 5 years old. um the way that these different areas or regions of the brain kind of interact with one another and we can think about it as a form of you know that's what that's what all these kind of wildl looking formalisms are for as a bunch of kind of um message passing schemes between these parts of the brain themselves operating at different scales. That's what allows for a lot of these more intricate kind of uh deep uh both temporally deep and then just kind of computationally deep uh you know broader systems that make up you know what is one's brain if not many many smaller parts. They're all interacting with one another um in different ways and sending different kinds of signals to one another. So, um, someone could reflect about what's happened to them over this past week, which will both bring up memories of today and things that are currently in short-term memory. It will also start to retrieve from that hippocample structure I mentioned a moment ago. um um kind of longer term memory of what happened you know say last uh Thursday or whatever it was and start to kind of construct through observing one's thoughts as it were hidden states of the world including for this moment like what does it mean to reflect on this past week what was going on this past week what do I make of that is there some kind of meaning could to to to consolidate and condense out of that um you It's we it's ultimately a kind of belief uh structured there. So I I also recognize we kind of ran up on on time here. Uh so we will have to have to end. But uh I uh team and Carrie, I really appreciate your folks questions. Um appreciate the engagement. uh next week at this time um this exact time next week we'll be talking about the Daniel and I do this kind of dovetailing back and forth between cohorts and chapters. So it'll it'll be me here at this time next week, but we'll be uh talking about I believe chapter 8 um in the textbook which is on continuous time models. um that will have to do with some interesting things that have to do with attractors. It'll have to do with some things that relate to this kind of message passing stuff. But it'll be it'll be really focused on some other ideas too of like there's a concept called generalized synchrony which is what happens whenever you could use it to model like communication for example. What does it mean for two agents to interact with one another? And what does it mean whenever we start to realize that like it's more commonly than not their beliefs are more likely to kind of synchronize over time as they continue to communicate. Um, which is found in humans. Even this the speed at which we speak tends to modulate to one another whenever we're in conversation with one another or we start to borrow idioms and things. But that's an empirical example. But anyway, so it's it's a different kind of chapter was rather rather interesting. Um and uh yeah, so hope to to see me next week. Drop in to that one. Drop into the morning one. Um any questions regardless of what chapter it is, uh Daniel and I will be happy to to discuss them. So um all right. Thanks so much everyone. Take care. >> Thanks. Take care. Thanks.